Product Requirements Document (PRD): BhumiSetu Land Case File & Property Intelligence Report

1. Document Overview & Product Vision

Product Name: BhumiSetu Land Case File (LCF) / Land Intelligence Report

Vision:
Transform fragmented, unstructured land records including revenue documents, CAD maps, field survey data, and legal opinions into a single, unified case file. This deliverable is designed to provide decision confidence to land buyers, owners, and developers.

Core Value Proposition:
Remove the "customer as project manager" burden by synthesizing Paper Reality, Spatial Reality, and Ground Reality into a clear, actionable risk assessment.

2. Target Persona & Key Use Cases

- **Land Buyer:** Seeking a "Verify First, Commit Later" decision checkpoint prior to committing capital.
- **Remote Owner:** Requiring asset visibility, boundary verification, and encroachment monitoring.
- **Developer / Institutional Investor:** Needing land intelligence, development potential, and spatial risk analysis.
- **Family / Inherited Property:** Seeking legal clarity, record organization, and boundary reconciliation among multiple claimants.

3. Core Report Architecture & Module Breakdown

Module 1: Executive Summary & Property Identity

This module serves as the initial snapshot of the property.

- **Property ID & Unique BhumiSetu Handle:** Unique identifier for the case file.
- **Property Location & Geolocation:** GPS coordinates, Village/Mouza, Thana No., Anchal, District, State.
- **Transaction Overview:** Offered size, asking price, and target land use (Residential / Commercial / Agricultural).
- **BhumiSetu Risk Score & Decision Badge:** Visual status indicator:
    - **Green:** Low Risk / Proceed.
    - **Amber:** Conditional / Specific Actions Required.
    - **Red:** High Risk / Major Discrepancies.
- **Summary Verdict:** A 3-4 sentence plain-language synthesis of the property status.

Module 2: Document Verification Matrix (Paper Reality)

A detailed audit of the legal and revenue documentation.

- **Document Ledger:** List of ingested documents including Sale Deed, Khatian / Khatiyan, Jamabandi / Register II, Mutation Order, Tax Receipts, and Encumbrance Certificate / LPC.
- **Ownership Chain Tracking:** Verified sequence of transfers and title continuity covering a 30+ year period.
- **Revenue Record Match:** Verification against official state revenue portals (e.g., Bihar Bhumi / Land Records).
- **Discrepancy Flags:** Highlighting issues in owner name spellings, share calculation errors, un-mutated portions, or pending litigations.

Module 3: Spatial & Boundary Analysis (Spatial Reality)

An analysis of the physical dimensions versus recorded data.

- **Revenue Map (Khasra Map) Overlay:** Geo-referenced digitization of official revenue maps.
- **Survey & GIS Mapping:** High-accuracy boundary plots using DGPS, Drone, or Total Station data.
- **Area Comparison Triangulation:**
    - Deed Area vs. Revenue Record Area vs. Physical Measured Area.
- **Boundary & Shape Analysis:** Frontage width, plot depth, access road alignment, and shape regularity.
- **Encroachment & Overlay Detection:** Overlap analysis with adjacent plots, government land, water bodies (Sairat/Khas Mahal), or ROW easements.

Module 4: Field Inspection & Ground Verification (Ground Reality)

Observations from physical on-site visits.

- **Physical Possession & Boundary Status:** Status of fencing, pillar markers, standing structures, and agricultural activity.
- **Access & Approach Road:** Verification of physical road width compared to recorded road right-of-way.
- **Local Inquiry & Possession Check:** On-ground insights regarding local dispute history, oral claims, or third-party possession.
- **Topography & Soil Environment:** Land level relative to the road, flooding risk, and proximity to high-tension power lines or pipelines.

Module 5: Integrated Risk & Gap Matrix

A consolidated view of identified risks.

Risk Category	Risk Level (High/Medium/Low)
Title & Ownership Risk	Person

Risk Category	Risk Level (High/Medium/Low)
Revenue & Mutation Gap	Person
Spatial & Boundary Discrepancy	Person
Ground & Access Risk	Person

- **Unresolved Gaps:** Specific missing documents or unverified claims requiring specialist follow-up.

Module 6: Recommended Next Action Plan

Step-by-step checklist for the customer.

- **Immediate Actions:** e.g., Request specific revenue correction, demand mutation proof from seller.
- **Negotiation Leverage:** e.g., Price adjustment recommendations based on area deficits.
- **Professional Referrals:** e.g., Targeted legal notice drafts or re-survey requests.

4. Technical Ingestion & Data Workflow

- **Input Sources:**
  - Customer uploads (PDF/Images).
  - Surveyor inputs (DXF/GeoJSON/KML).
  - Field Inspector checklist via Mobile App.
  - Legal Advisor inputs via structured forms.
- **Synthesis Engine:** Automated cross-referencing between Deed text fields, CAD vector boundaries, and revenue database extracts.
- **Deliverable Formats:**
  - Interactive Web Dashboard (featuring GIS viewer, document viewer, and collapsible modules).
  - Downloadable Executive Summary in PDF format.

5. Success Metrics & Quality Standards

- **Turnaround Time (TAT):** Report generation must be completed within 48-72 hours of field survey completion.
- **Accuracy Guarantee:** 100% spatial reconciliation accuracy between measured and paper boundaries.
- **Customer Decision Clarity Score:** Measured via post-delivery survey ratings on decision confidence.